ITA0605 - Machine Learning
List of Lab Exercises

16. Compare different types Classification Algorithms and evaluate their performance.

Aim:

To compare different classification algorithms and evaluate their performance on a given dataset.

Algorithm:

1. Import the required Python libraries.
2. Load the classification dataset.
3. Split the dataset into training and testing sets.
4. Train the Logistic Regression, Decision Tree, KNN, SVM, Naive Bayes, and Random Forest models.
5. Predict the test data using all six classification algorithms.
6. Compare their performance using accuracy, precision, recall, F1-score, and display the results.

Dataset 1:

Program Code:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score

data = {
    'CGPA': [9.2, 8.8, 8.5, 7.2, 6.8, 9.0, 7.5, 8.6, 6.5, 8.9],
    'Aptitude': [90, 85, 80, 72, 68, 92, 75, 84, 60, 88],
    'Communication': [88, 82, 79, 70, 65, 90, 74, 81, 58, 86],
    'Placement': ['Placed', 'Placed', 'Placed', 'Not Placed', 'Not Placed',
                  'Placed', 'Not Placed', 'Placed', 'Not Placed', 'Placed']
}

df = pd.DataFrame(data)
X = df[['CGPA', 'Aptitude', 'Communication']]
y = df['Placement']
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42, stratify=y
)
dt_model = DecisionTreeClassifier(random_state=42)
dt_model.fit(X_train, y_train)
```

```
knn_model = KNeighborsClassifier(n_neighbors=3)
knn_model.fit(X_train, y_train)
dt_pred = dt_model.predict(X_test)
knn_pred = knn_model.predict(X_test)
dt_accuracy = accuracy_score(y_test, dt_pred)
knn_accuracy = accuracy_score(y_test, knn_pred)

print("Decision Tree Accuracy:", dt_accuracy)
print("KNN Accuracy:", knn_accuracy)
new_sample = [[8.7, 86, 84]]

print("\nNew Sample Prediction:")
print("Decision Tree:", dt_model.predict(new_sample)[0])
print("KNN:", knn_model.predict(new_sample)[0])
```

Output:

```
print("\nNew Sample Prediction:")
print("Decision Tree:", dt_model.predict(new_sample)[0])
print("KNN:", knn_model.predict(new_sample)[0])

Decision Tree Accuracy: 1.0
KNN Accuracy: 1.0

New Sample Prediction:
Decision Tree: Placed
KNN: Placed
```

Dataset 2:

Program Code:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score
data = {
    'Income': [8, 7, 6, 4, 3, 9, 5, 8, 4, 7],
    'CreditScore': [780, 760, 730, 620, 590, 810, 650, 770, 610, 750],
    'LoanAmount': [5, 4, 5, 8, 9, 4, 7, 5, 8, 6],
    'LoanStatus': ['Approved', 'Approved', 'Approved', 'Rejected', 'Rejected',
                   'Approved', 'Rejected', 'Approved', 'Rejected', 'Approved']
}

df = pd.DataFrame(data)

X = df[['Income', 'CreditScore', 'LoanAmount']]
y = df['LoanStatus']
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

dt = DecisionTreeClassifier(random_state=42)
dt.fit(X_train, y_train)
knn = KNeighborsClassifier(n_neighbors=3)
knn.fit(X_train, y_train)

dt_pred = dt.predict(X_test)
knn_pred = knn.predict(X_test)

dt_acc = accuracy_score(y_test, dt_pred)
knn_acc = accuracy_score(y_test, knn_pred)

print("Decision Tree Accuracy:", dt_acc)
print("KNN Accuracy:", knn_acc)
new_sample = [[7, 760, 5]]

print("\nNew Sample Prediction:")
print("Decision Tree:", dt.predict(new_sample)[0])
print("KNN:", knn.predict(new_sample)[0])
```

Output:

```
# New sample
new_sample = [[7, 760, 5]]

print("\nNew Sample Prediction:")
print("Decision Tree:", dt.predict(new_sample)[0])
print("KNN:", knn.predict(new_sample)[0])
```

```
... Decision Tree Accuracy: 1.0
    KNN Accuracy: 1.0

    New Sample Prediction:
    Decision Tree: Approved
    KNN: Approved
```

Dataset 3:

Program Code:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score
data = {
    'Temperature': [39.0, 38.5, 37.0, 39.2, 36.8, 38.8, 37.2, 39.1, 36.7, 38.9],
    'HeartRate': [110, 108, 78, 115, 75, 112, 80, 114, 74, 111],
```

```
    'OxygenLevel': [94, 95, 99, 93, 98, 94, 98, 92, 99, 93],
    'Disease': ['Positive', 'Positive', 'Negative', 'Positive', 'Negative',
                'Positive', 'Negative', 'Positive', 'Negative', 'Positive']
}

df = pd.DataFrame(data)

X = df[['Temperature', 'HeartRate', 'OxygenLevel']]
y = df['Disease']
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

dt = DecisionTreeClassifier(random_state=42)
dt.fit(X_train, y_train)

knn = KNeighborsClassifier(n_neighbors=3)
knn.fit(X_train, y_train)
dt_pred = dt.predict(X_test)
knn_pred = knn.predict(X_test)

dt_acc = accuracy_score(y_test, dt_pred)
knn_acc = accuracy_score(y_test, knn_pred)

print("Decision Tree Accuracy:", dt_acc)
print("KNN Accuracy:", knn_acc)

new_sample = [[38.7, 109, 94]]

print("\nNew Sample Prediction:")
print("Decision Tree:", dt.predict(new_sample)[0])
print("KNN:", knn.predict(new_sample)[0])
```

Output:

```
new_sample = [[38.7, 109, 94]]

print("\nNew Sample Prediction:")
print("Decision Tree:", dt.predict(new_sample)[0])
print("KNN:", knn.predict(new_sample)[0])
```

```
Decision Tree Accuracy: 1.0
KNN Accuracy: 1.0

New Sample Prediction:
Decision Tree: Positive
KNN: Positive
```

Dataset 4:

Program Code:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score
data = {
    'Experience': [10, 8, 7, 3, 2, 9, 4, 8, 2, 6],
    'Performance': [95, 90, 88, 65, 60, 93, 70, 89, 58, 85],
    'TrainingHours': [50, 45, 40, 20, 18, 48, 25, 42, 15, 38],
    'Promotion': ['Promoted', 'Promoted', 'Promoted', 'Not Promoted',
                  'Not Promoted', 'Promoted', 'Not Promoted', 'Promoted',
                  'Not Promoted', 'Promoted']
}

df = pd.DataFrame(data)

X = df[['Experience', 'Performance', 'TrainingHours']]
y = df['Promotion']
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)
dt = DecisionTreeClassifier(random_state=42)
dt.fit(X_train, y_train)
knn = KNeighborsClassifier(n_neighbors=3)
knn.fit(X_train, y_train)
dt_pred = dt.predict(X_test)
knn_pred = knn.predict(X_test)
dt_acc = accuracy_score(y_test, dt_pred)
knn_acc = accuracy_score(y_test, knn_pred)

print("Decision Tree Accuracy:", dt_acc)
print("KNN Accuracy:", knn_acc)
new_sample = [[7, 90, 44]]

print("\nNew Sample Prediction:")
print("Decision Tree:", dt.predict(new_sample)[0])
print("KNN:", knn.predict(new_sample)[0])
```

Output:

```
print("\nNew Sample Prediction:")
print("Decision Tree:", dt.predict(new_sample)[0])
print("KNN:", knn.predict(new_sample)[0])
```

```
Decision Tree Accuracy: 1.0
KNN Accuracy: 1.0

New Sample Prediction:
Decision Tree: Promoted
KNN: Promoted
```

Dataset 5:

Program Code:

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score
data = {
    'EngineCapacity': [100, 125, 150, 800, 1000, 1200, 110, 900, 140, 1300],
    'Weight': [95, 110, 125, 850, 950, 1050, 100, 900, 120, 1100],
    'Mileage': [70, 65, 55, 22, 20, 18, 68, 21, 58, 17],
    'VehicleType': ['Bike', 'Bike', 'Bike', 'Car', 'Car', 'Car',
                    'Bike', 'Car', 'Bike', 'Car']
}

df = pd.DataFrame(data)

X = df[['EngineCapacity', 'Weight', 'Mileage']]
y = df['VehicleType']

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)
dt = DecisionTreeClassifier(random_state=42)
dt.fit(X_train, y_train)

knn = KNeighborsClassifier(n_neighbors=3)
knn.fit(X_train, y_train)

dt_pred = dt.predict(X_test)
knn_pred = knn.predict(X_test)
```

```
dt_acc = accuracy_score(y_test, dt_pred)
knn_acc = accuracy_score(y_test, knn_pred)

print("Decision Tree Accuracy:", dt_acc)
print("KNN Accuracy:", knn_acc)

new_sample = [[135, 118, 60]]

print("\nNew Sample Prediction:")
print("Decision Tree:", dt.predict(new_sample)[0])
print("KNN:", knn.predict(new_sample)[0])
```

Output:

```
print("\nNew Sample Prediction:")
print("Decision Tree:", dt.predict(new_sample)[0])
print("KNN:", knn.predict(new_sample)[0])
```

```
Decision Tree Accuracy: 1.0
KNN Accuracy: 1.0

New Sample Prediction:
Decision Tree: Bike
KNN: Bike
```

Result:

Decision Tree and KNN algorithms were successfully applied to all five datasets.
The accuracy of both algorithms was calculated and compared.
The algorithm with the higher accuracy performed better.